# Fabio Miranda

Assistant Professor, Department of Computer Science, College of Engineering
University of Illinois Chicago
*851 S. Morgan St, MC 152, Chicago, IL, 60607*
fabiom@uic.edu | fmiranda.me

## Research Overview

I am interested in developing techniques that allow for the interactive visual analysis of large-scale data, combining methods from visualization, data management, AI, and computer graphics. I have worked closely with domain experts from different fields and the outcome of these collaborations included not only research published in leading venues, but also systems that were made available to experts in academia, industry and government agencies. My work has also received extensive coverage from different media outlets, including The New York Times, The Economist, Architectural Digest, among others. Over the past five years, I have secured more than $1.9 million in funding (my share) from NSF, NIH, and DOT, published more than 20 papers in top-tier computer science conferences and journals, and released several open, accessible software tools with more than 500 GitHub stars across repositories. My work has been recognized with several honors, including the SIGMOD Best Demo Award and an IEEE VIS Best Paper Honorable Mention. I also serve regularly on the IEEE VIS Program Committee and will serve as General Chair of IEEE VIS 2027.

## Education

2012 - 2018 **Ph.D. in Computer Science** — New York, NY, USA
New York University (NYU)
Advised by Professor Cláudio T. Silva, IEEE Fellow
Dissertation: "Data structures for the interactive visual analysis of urban data"

2009 - 2011 **M.Sc. in Computer Science** — Rio de Janeiro, RJ, Brazil
Pontifical Catholic University of Rio de Janeiro (PUC-Rio)
Advised by Professor Waldemar Celes
Thesis: "Volume rendering of unstructured hexahedral meshes"

2005 - 2009 **B.Sc. in Computer Science** — Belo Horizonte, MG, Brazil
Federal University of Minas Gerais (UFMG)
Advised by Professor Luiz Chaimowicz

## Professional Experience

Oct. 2020 - present **University of Illinois Chicago** — Chicago, IL, USA
Assistant Professor, Department of Computer Science, College of Engineering

Fall 2018 - Fall 2020 **New York University** — New York, NY, USA
Postdoctoral researcher

Summer 2016 **Argonne National Laboratory** — Lemont, IL, USA
Research intern, Mentor: Venkatram Vishwanath

Summer 2015 **IBM T.J. Watson Research Center** — Yorktown Heights, NY, USA
Research intern, Mentor: Bruce D'Amora

Summer 2014 **AT&T Research** — Middletown, NJ, USA
Research intern, Mentors: Lauro Lins and James Klosowski

Summer 2013 **Sandia National Laboratories** — Albuquerque, NM, USA
Research intern, Mentor: Patricia Crossno

Fall 2012 - Fall 2018 **New York University** — New York, NY, USA
Research assistant

2009 - 2012 **TecGraf / PUC-Rio** — Rio de Janeiro, Brazil
Research assistant

## Awards & Honours

2023 IEEE VIS 2023 Best Paper Honorable Mention
For "ProWis: A Visual Approach for Building, Managing, and Analyzing Weather Simulation Ensembles at Runtime".

2023 SIBGRAPI 2023 Best Paper Honorable Mention
For "Visual Analytics for Profiling Land Use Changes".

2022 National Academy of Sciences Kavli Fellow
Cross-disciplinary program recognizing exceptional early-career scientists.

2018 SIGMOD Best Demonstration Award
For "Interactive Visual Exploration of Spatio-Temporal Urban Data Sets Using Urbane".

2018 Pearl Brownstein Doctoral Research Award
For doctoral research that shows the greatest promise, awarded by NYU.

2010-2012 CAPES and Petrobras Fellowships
Awarded during M.S. studies.

2006-2009 FINEP and CNPq Fellowships
Awarded during B.S. studies.

## Selected Media Coverage

November 2022 A scrap metal shredder on the Southwest Side wants to renew its permit. But Pilsen residents are fighting back.
Chicago Tribune

February 2022 Escuelas y comunidades latinas en Chicago son las más afectadas por la contaminación, según estudio.
Univision Chicago

November 2017 Urban Pulse maps, analyzes use of urban spaces.
GCN

September 2017 Urban Pulse Uses Social Media Data to Show Cities in a New Light.
Architectural Digest

September 2017 New program wants to improve cities with the power of tweets and Flickr uploads.
Curbed

December 2016 Mapping the Shadows of New York City: Every Building, Every Block.
The New York Times

October 2016 Listen to the music of the traffic in the city.
The Economist

## Publications

Underlined name: advised student
J: journal, C: conference / symposium, A: abstract, W: workshop
Author order: in computer science, and in visualization and human-computer interaction in particular, the first author is the primary contributor (typically the lead student), and the senior author is customarily listed last
Note: IEEE VIS, IEEE VR, and EuroVis full papers undergo full journal-style peer review and are published as regular articles in IEEE Transactions on Visualization and Computer Graphics, and Computer Graphics Forum journals.

Accepted:

[C27] 2026 Occlusion-Free Conformal Lensing for Spatiotemporal Visualization in 3D Urban Analytics
R. Mota, J. D. Silva, **F. Miranda**, U. R. Alim, E. Sharlin, N. Ferreira
*IEEE Transactions on Visualization and Computer Graphics, vol. 32, no. 5, 2026, pp. 4174-4184*
(IEEE VR 2026, Full paper)

[J17] 2026 A Neural Field-Based Approach for View Computation & Data Exploration in 3D Urban Environments
S. Cobeli, K. Omar, R. Valenca, N. Ferreira, **F. Miranda**
*IEEE Transactions on Visualization and Computer Graphics, vol. 32, no. 2, 2026, pp. 1540-1553*

[C26] 2026 Urbanite: A Dataflow-Based Framework for Human-AI Interactive Alignment in Urban Visual Analytics
G. Moreira, L. Ferreira, C. Veiga, M. Hosseini, **F. Miranda**
*IEEE Transactions on Visualization and Computer Graphics, vol. 32, no. 1, Jan 2026, pp. 1065-1075*
(IEEE VIS 2025, Full paper)

[C25] 2026 VA-Blueprint: Uncovering Building Blocks for Visual Analytics System Design
L. Ferreira, G. Moreira, **F. Miranda**
*IEEE Transactions on Visualization and Computer Graphics, vol. 32, no. 1, Jan 2026, pp. 1197-1207*
(IEEE VIS 2025, Full paper)

[C24] 2026 StreetWeave: A Declarative Grammar for Street-Overlaid Visualization of Multivariate Data
S. Srabanti, G. E. Marai, **F. Miranda**
*IEEE Transactions on Visualization and Computer Graphics, vol. 32, no. 1, Jan 2026, pp. 1054-1064*
(IEEE VIS 2025, Full paper)

[C23] 2025 Real-Time Monitoring and Historical Analysis of Rainfall and its Impacts on Urban Areas
F. V. da Cruz, **F. Miranda**, A. Paes, D. de Oliveira, M. Lage
*38th Conference on Graphics Patterns and Images (SIBGRAPI 2025), pp. 1-6*
(SIBGRAPI 2025, Full paper)

[C22] 2025 Towards Enabling the Analysis of Visual Exploration Processes through Interaction Provenance
L. S. de Oliveira, G. Moreira, **F. Miranda**, M. Lage, D. de Oliveira
*40th Brazilian Symposium on Databases (SBBD 2025), pp. 858-864*

[W6] 2025 Immersive Analytics at the Electronic Visualization Laboratory
L. Renambot, J. Talandis, **F. Miranda**, L. Long, D. J. Sandin, G. E. Marai, M. E. Papka, N. Soni, A. E. Johnson, T. A. DeFanti, D. Tsoupikova, D. Plepys, M. D. Brown, J. Leigh
*2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), pp. 206-210*
(IEEE VR 2025, Workshop paper)

[J16] 2025 VIGMA: An Open-Access Framework for Visual Gait and Motion Analytics
K. Omar, S. Wang, R. Kungumaraju, T. Bhatt, **F. Miranda**
*IEEE Transactions on Visualization and Computer Graphics, vol. 31, no. 10, 2025, pp. 8143-8158*

[J15] 2025 Automatic gait event detection in older adults during perturbed walking
S. Wang, K. Omar, **F. Miranda**, T. Bhatt
*Journal of NeuroEngineering and Rehabilitation, 22, 2025, pp. 1-10*

[C21] 2025 Curio: A Dataflow-Based Framework for Collaborative Urban Visual Analytics
G. Moreira, M. Hosseini, C. Veiga, L. Alexandre, N. Colaninno, D. de Oliveira, N. Ferreira, M. Lage, **F. Miranda**
*IEEE Transactions on Visualization and Computer Graphics, vol. 31, no. 1, Jan 2025, pp. 1224-1234*
(IEEE VIS 2024, Full paper)

[J14] 2025 Deep Umbra: A Generative Approach for Sunlight Access Computation in Urban Spaces
K. Omar, G. Moreira, D. Hodczak, M. Hosseini, N. Colaninno, M. Lage, **F. Miranda**
*IEEE Transactions on Big Data, vol. 11, no. 2, 2025, pp. 388-401*

[C20] 2024 The Future of Urban Accessibility: The Role of AI
J. Froehlich, C. Li, M. Hosseini, **F. Miranda**, A. Sevtsuk, Y. Eisenberg
*ACM SIGACCESS Conference on Computers and Accessibility, 2024, pp. 1-6*
(ACCESS'24, Short paper)

[J13] 2024 Assessing the landscape of toolkits, frameworks, and authoring tools for urban visual analytics systems
L. Ferreira, G. Moreira, M. Hosseini, M. Lage, N. Ferreira, **F. Miranda**
*Computers & Graphics, vol. 123, 2024, pp. 1-17*

[C19] 2024 A Survey on Visual Analytics for 3D Urban Data
**F. Miranda**, T. Ortner, G. Moreira, M. Hosseini, F. Biljecki, C. T. Silva, M. Lage, N. Ferreira
*Computer Graphics Forum, vol. 43, no. 3, 2024, pp. 1-31*
Top-10 most cited CGF papers in 2025 (EuroVis 2024, Full paper)

[C18] 2024 The Urban Toolkit: A Grammar-based Framework for Urban Visual Analytics
G. Moreira, M. Hosseini, M.N.A. Nipu, M. Lage, N. Ferreira, **F. Miranda**
*IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 1, Jan 2024, pp. 1402-1412*
(IEEE VIS 2023, Full paper)

[C17] 2024 ProWis: A Visual Approach for Building, Managing, and Analyzing Weather Simulation Ensembles at Runtime
C.V.F. de Souza, S.M. Bonnet, D. de Oliveira, M. Cataldi, **F. Miranda**, M. Lage
*IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 1, Jan 2024, pp. 738-747*
Best Paper Honorable Mention (IEEE VIS 2023, Full paper)

[C16] 2023 Visual Analytics for Profiling Land Use Changes
C. Santos, M. Hosseini, J. Rulff, **F. Miranda**, L. Wilson, C. Silva, N. Ferreira, M. Lage
*36th Conference on Graphics Patterns and Images (SIBGRAPI 2023), pp. 1-6*
Best Paper Honorable Mention (SIBGRAPI 2023, Full paper)

[C15] 2023 Visual Analytics Using Heterogeneous Urban Data
S. Bonadia, R. Gama, D. Oliveira, **F. Miranda**, M. Lage
*36th Conference on Graphics Patterns and Images (SIBGRAPI 2023), pp. 1-6*
(SIBGRAPI 2023, Full paper)

[J12] 2023 Environmental Justice through Community-Policy Participatory Partnerships
P.A. Boda, F. Fusi, **F. Miranda**, G.M.M. Palmer, J. Flax-Hatch, M. Siciliano, A. Sambanis, L. Johnson, S. Derrible, M. Cailas
*Journal of Environmental Protection, vol. 14, no. 8, Aug 2023, pp. 616-636*

[C13] 2023 A Comparison of Spatiotemporal Visualizations for 3D Urban Analytics
R. Mota, M. Horga, N. Ferreira, J. D. Silva, M. Lage, L. Ceferino, U. R. Alim, E. Sharlin, **F. Miranda**
*IEEE Transactions on Visualization and Computer Graphics, vol. 29, no. 1, Jan 2023, pp. 1277-1287*
(IEEE VIS 2022, Full paper)

[C14] 2023 Does a Quieter City Mean Fewer Complaints? The Sounds of New York City During COVID-19 Lockdown
M. Cartwright, M. Fuentes, C. Mydlarz, **F. Miranda**, J. P. Bello
*IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2023), pp. 1-5*
(ICASSP, Full paper)

[J11] 2023 Mapping the Walk: A Scalable Computer Vision Approach for Generating Sidewalk Network Datasets
M. Hosseini, A. Sevtsuk, **F. Miranda**, R. M. Cesar Jr, C. T. Silva
*Computers, Environment and Urban Systems, vol. 101, April 2023, pp. 1-22*
Featured on MIT Press

[A2] 2022 Interactive Visual Analysis of Urban Data: Applications in the Weather and Climate Domains
**F. Miranda**
*American Geophysical Union Fall Meeting 2022*

[A1] 2022 Mapping Sidewalk Fall Risks Using Big Data And Machine Learning
**F. Miranda**, M. Hosseini
*Innovation in Aging, vol. 6, Supplement 1, Nov 2022*

[W5] 2022 Crowdsourcing and Sidewalk Data: A Preliminary Study on the Trustworthiness of OpenStreetMap Data in the US
K. Omar, G. Moreira, D. Hodczak, M. Hosseini, **F. Miranda**
*ASSETS 2022 UrbanAccess Workshop, pp. 1-7*

[C12] 2022 The Future of Urban Accessibility for People with Disabilities: Data Collection, Analytics, Policy, and Tools
J. Froehlich, Y. Eisenberg, M. Hosseini, **F. Miranda** et al.
*ACM SIGACCESS Conference on Computers and Accessibility, 2022, pp. 1-8*
(ASSETS '22, Short paper)

[J10] 2022 A Comparative Study of Methods for Visualization of Probability Distributions of Geographical Data
S. Srabanti, C. V. de Souza, E. J. da Silva, M. Lage, N. Ferreira, **F. Miranda**
*Multimodal Technologies and Interaction 6 (7), 53, 2022, pp. 1-20*

[W4] 2022 Towards Global-Scale Crowd+AI Techniques to Map and Assess Sidewalks for People with Disabilities
M. Hosseini, M. Saugstad, **F. Miranda**, A. Sevtsuk, C. T. Silva, J. E. Froehlich
*AVA: Accessibility, Vision, and Autonomy Meet, pp. 1-6*
(CVPR 2022 Workshop)

[J9] 2022 Near-fall detection in unexpected slips during over-ground locomotion
S. Wang, **F. Miranda**, Y. Wang, R. Rasheed, T. Bhatt
*Sensors, vol. 22, no. 9, pp. 1-13*

[C11] 2022 Urban Rhapsody: Large-scale Visual Exploration of Urban Soundscapes
J. Rulff, **F. Miranda**, M. Hosseini, M. Lage, M. Cartwright, G. Dove, J. P. Bello, C. Silva
*Computer Graphics Forum, vol. 41, no. 3, Jun 2022, pp. 209-221*
(Eurovis 2022, Full paper)

[C10] 2022 A Tale of Two Centers: Visual Exploration of Health Disparities in Cancer Care
S. Srabanti, M. Tran, V. Achim, D. Fuller, G. Canahuate, **F. Miranda**, G.E. Marai
*IEEE Pacific Visualization Symposium, 2022, pp. 1-10*
(PacificVis 2022, Full paper)

[J8] 2022 CitySurfaces: City-scale Semantic Segmentation of Sidewalks Surfaces
M. Hosseini, **F. Miranda**, J. Lin, C. Silva
*Sustainable Cities and Society, vol. 79, Apr 2022, pp. 1-13*

[J7] 2022 Visualizing Simulation Ensembles of Extreme Weather Events
C. V. de Souza, P. Luz, M. Cataldi, **F. Miranda**, M. Lage
*Computers & Graphics, vol. 104, May 2022, pp. 162-172*

[J6] 2022 UrbanRama: Navigating Cities in Virtual Reality
S. Chen, **F. Miranda**, N. Ferreira, M. Lage, H. Doraiswamy, C. Brenner, C. Defanti, M. Koutsoubis, L. Wilson, K. Perlin, C. Silva
*IEEE Transactions on Visualization and Computer Graphics, vol. 28, no. 12, Dec 2022, pp. 4685-4699*

[C9] 2021 Visualizing Environmental Justice Issues in Urban Areas with a Community Input Approach
J. Flax-Hatch, S. Srabanti, **F. Miranda**, A. Sambanis, M. Cailas
*2nd Spatial Data Science Symposium, pp. 1-6*
Featured on Univision Chicago

[C8] 2021 Sidewalk Measurements from Satellite Images: Preliminary Findings
M. Hosseini, I. B. Araujo, H. Yazdanpanah, E. Tokuda, **F. Miranda**, C. Silva, R. M. Cesar Jr
*2nd Spatial Data Science Symposium, pp. 1-6*

[W3] 2021 COVID-19 EnsembleVis: Visual Analysis of County-level Ensemble Forecast Models
S. Srabanti, G. E. Marai, **F. Miranda**
*12th Workshop on Visual Analytics in Healthcare, 2021, pp. 1-6*

[W2] 2021 Transportation Scenario Planning with Graph Neural Network
A. A. Peregrino, S. Pradhan, Z. Liu, N. Ferreira, **F. Miranda**
*10th International Workshop on Urban Computing, 2021, pp. 1-5*

[C7] 2020 Urban Mosaic: Visual Exploration of Streetscapes Using Large-scale Image Data
**F. Miranda**, M. Lage, H. Doraiswamy, M. Hosseini, G. Dove, C. T. Silva
*2020 CHI Conference on Human Factors in Computing Systems, pp. 1-15*

[C6] 2020 Learning Geo-Contextual Embeddings for Commuting Flow Prediction
Z. Liu, **F. Miranda**, W. Xiong, J. Yang, Q. Wang, C. T. Silva
*Thirty-Fourth AAAI Conference on Artificial Intelligence, vol. 34, no. 01, pp. 808-816*

[J5] 2019 Shadow Accrual Maps: Efficient Accumulation of City-Scale Shadows over Time
**F. Miranda**, H. Doraiswamy, M. Lage, L. Wilson, M. Hsieh, C. T. Silva
*IEEE Transactions on Visualization and Computer Graphics, vol. 25, no. 3, pp. 1559-1574, Mar 2019*
Featured on The New York Times

[C5] 2018 Time Lattice: A Data Structure for the Interactive Visual Analysis of Large Time Series
**F. Miranda**, M. Lage, H. Doraiswamy, C. Mydlarz, J. Salamon, Y. Lockerman, J. Freire, C. T. Silva
*Computer Graphics Forum, vol. 37, no. 3, pp. 23-35, Jun 2018*
(EuroVis 2018, Full paper)

[C4] 2018 Interactive Visual Exploration of Spatio-Temporal Urban Data Sets using Urbane
H. Doraiswamy, E. Tzirita Zacharatou, **F. Miranda**, M. Lage, A. Ailamaki, C. T. Silva, J. Freire
*2018 ACM SIGMOD Intl. Conf. on Management of Data - Demo, pp. 1693-1696*
Best Demonstration Award

[J4] 2018 Spatio-Temporal Urban Data Analysis: A Visual Analytics Perspective
H. Doraiswamy, J. Freire, M. Lage, **F. Miranda**, C. T. Silva
*IEEE Computer Graphics and Applications, vol. 38, no. 5, pp. 26-35, Sept/Oct 2018*

[J3] 2018 TopKube: A Rank-Aware Data Cube for Real-Time Exploration of Spatiotemporal Datasets
**F. Miranda**, L. Lins, J. Klosowski, C. T. Silva
*IEEE Transactions on Visualization and Computer Graphics, vol. 24, no. 3, pp. 1394-1407, Mar 2018*

[J2] 2017 Urban Pulse: Capturing the Rhythm of Cities
**F. Miranda**, H. Doraiswamy, M. Lage, K. Zao, B. Goncalves, L. Wilson, M. Hsieh, C. T. Silva
*IEEE Transactions on Visualization and Computer Graphics, vol. 23, no. 1, pp. 791-800, Jan 2017*
Featured on The Economist, invited to SIGGRAPH 2017 TVCG special session

[C3] 2017 Data Visualization Tool for Monitoring Transit Operation and Performance
A. Kurkcu, **F. Miranda**, K. Ozbay, C. T. Silva
*5th IEEE Intl. Conf. on Models and Technologies for Intelligent Transportation Systems (2017), pp. 598-603*

[W1] 2016 TopKube: A Rank-Aware Data Cube for Real-Time Exploration of Spatiotemporal Datasets
**F. Miranda**, L. Lins, J. Klosowski, C. T. Silva
*Data Systems for Interactive Analysis (DSIA) 2016*

[J1] 2012 Volume Rendering of Unstructured Hexahedral Meshes
**F. Miranda**, and W. Celes
*The Visual Computer Journal, vol. 28, no. 10, pp. 1005-1014, Oct 2012*

[C2] 2011 Accurate Volume Rendering of Unstructured Hexahedral Meshes
**F. Miranda**, and W. Celes
*24th Sibgrapi Conference on Graphics, Patterns and Images (2011), pp. 93-100*

[C1] 2011 Illustrative Volume Visualization for Unstructured Meshes Based on Photic Extremum Lines
A. Rocha, **F. Miranda**, and W. Celes
*24th Sibgrapi Conference on Graphics, Patterns and Images (2011), pp. 101-108*

## Selected Open-Source Tools & Datasets

Recent tools are developed as part of the Urban Toolkit ecosystem

2026 Autark
Serverless toolkit for the rapid prototyping of urban visual analytics systems; successor to the Urban Toolkit (UTK).

2025 Urbanite

Dataflow-based framework for human-AI interactive alignment in urban visual analytics, integrated with Curio.

2025 StreetWeave

Declarative grammar for street-overlaid visualization of multivariate data.

2025 VIGMA

Open-access framework for visual gait and motion analytics, integrating computational notebooks and a Python library.

2025 VA-Blueprint

Structured, machine-readable knowledge base of the building blocks of visual analytics systems.

2025 Neural Field View Computation

Neural field-based method for view computation and data exploration in 3D urban environments.

2024 Curio

Dataflow-based framework for collaborative urban visual analytics.

2024 Deep Umbra & Global Shadow Dataset

Generative approach for sunlight access computation, with accumulated shadow data for over 100 cities across six continents.

2023 The Urban Toolkit (UTK)

Grammar-based framework for authoring web-based urban visualizations.

2019 New York City Shadow Data

Shadow data for New York City, also used by The New York Times.

2018 Urban Pulse

Open-source implementation of the Urban Pulse framework.

2017 Bus Explorer

Tool for exploring large-scale bus tracking data, developed in collaboration with the New York City Department of Transportation.

## External Grants (Share of awarded grants: $1,966,156)

Personal share calculated using the F&A split submitted to OVCR

Awarded:

Fall 2026 - Fall 2029 — **Visualizing Infrastructures for Vigorous Environmental Science in Chicago Public Schools** — The Spencer Foundation

Role: Co-PI

UIC's share: $374,298. Miranda's share: $93,575 (25% of UIC's share)

Fall 2024 - Fall 2029 — **Collaborative Research: Frameworks: Cyberinfrastructure to Catalyze and Sustain the Urban Computing Community** — NSF (Cyberinfrastructure for Sustained Scientific Innovation)

Role: UIC PI

UIC's share: $1,750,000. Miranda's share: $1,225,000 (70% of UIC's share)

Fall 2024 - Fall 2029 — **Aging and task-specific training to reduce falls** — NIH

Role: Co-I

UIC's share: $3,879,744. Miranda's share: $232,785 (6% of UIC's share)

Fall 2023 - Fall 2026 — **Research Infrastructure: MRI: Track 2 Acquisition of Data Observation and Computation Collaboratory (DOCC)** — NSF (Major Research Instrumentation Program)

Role: Co-PI

UIC's share: $1,548,545. Miranda's share: $154,854 (10% of UIC's share)

Fall 2023 - Fall 2028 | **Clean Energy and Equitable Transportation Solutions (CLEETS)** | NSF (US-UK Global Centers)

Role: SP
UIC's share: $1,181,443. Miranda's share: $177,216 (15% of UIC's share)

Fall 2023 - Fall 2025 | **Impact Measurement for At-Grade Crossings Analysis and Prioritization** | IDOT

Role: PI
Total: $263,634. UIC's share: $52,726. Miranda's share: $52,726 (20% of total share).

Fall 2022 - Fall 2023 | **Data readiness for 'Perturbation training for enhancing stability and limb support control for fall-risk reduction among stroke survivors'** | NIH

Role: Co-I
UIC's share: $308,918. Miranda's share: $30,000 (10% of UIC's share).

## Internal Grants

Fall 2025 - Fall 2027 | **SPILAB: Creating Pathways for Community-Centered Mobility in Illinois and São Paulo: Advancing Accessibility, Public Health, and Climate Resilience** | Brasillinois Collaborative Grant

Role: PI
UIC's share: $11,000

Fall 2025 - Fall 2026 | **Cracking the Big Data Challenge in Transportation** | University of Toronto – University of Illinois

Role: Co-PI
UIC's share: $11,000

Fall 2022 - Fall 2024 | **PRESUR: Planning a Resilient and Equitable State Using Real-time Data** | DPI

Role: Co-PI
UIC's share: $125,000. Miranda's share: $32,000 (25%)

## Teaching

400-level courses: Graduate and undergraduate courses
500-level courses: Graduate-only courses
Course evaluation scores are on a 0-5 scale; $n$ denotes the number of student responses

Spring 2026 | **CS524: Big Data Visualization & Analytics (Students: 22)** | UIC

Graduate session (n=14): Overall teaching effectiveness: 4.43; Overall quality of the course: 4.57
Course page

Fall 2025 | **CS424: Visualization & Visual Analytics (Students: 61)** | UIC

Graduate session (n=20): Overall teaching effectiveness: 4.5; Overall quality of the course: 4.6
Undergraduate session (n=7): Overall teaching effectiveness: 3.71; Overall quality of the course: 4.17
Course page

Spring 2025 | **CS424: Visualization & Visual Analytics (Students: 60)** | UIC

Graduate session (n=18): Overall teaching effectiveness: 4.67; Overall quality of the course: 4.61
Undergraduate session (n=11): Overall teaching effectiveness: 4.4; Overall quality of the course: 4.5
Course page

Fall 2024 | **CS524: Big Data Visualization & Analytics (Students: 30)** | UIC

Graduate session (n=27): Overall teaching effectiveness: 4.48; Overall quality of the course: 4.33
Course page

Spring 2024 **CS425: Computer Graphics I (Students: 39)** UIC
Graduate session (n=6): Overall teaching effectiveness: 4.83; Overall quality of the course: 4.83
Undergraduate session (n=26): Overall teaching effectiveness: 3.69; Overall quality of the course: 3.77
Course page

Fall 2023 **CS424: Visualization & Visual Analytics (Students: 90)** UIC
Graduate session (n=39): Overall teaching effectiveness: 4.7; Overall quality of the course: 4.73
Undergraduate session (n=40): Overall teaching effectiveness: 4.28; Overall quality of the course: 4.18
Course page

Spring 2023 **CS524: Big Data Visualization & Analytics (Students: 38)** UIC
Graduate session (n=35): Overall teaching effectiveness: 4.63; Overall quality of the course: 4.54
Course page

Fall 2022 **CS424: Visualization & Visual Analytics (Students: 50)** UIC
Graduate session (n=19): Overall teaching effectiveness: 4.63; Overall quality of the course: 4.58
Undergraduate session (n=22): Overall teaching effectiveness: 4.36; Overall quality of the course: 4.45
Course page

Spring 2022 **CS425: Computer Graphics I (Students: 49)** UIC
Graduate session (n=6). Overall teaching effectiveness: 4.67; Overall quality of the course: 4.67
Undergraduate session (n=31). Overall teaching effectiveness: 3.97; Overall quality of the course: 3.9
Course page

Fall 2021 **CS594: Big Data Visualization & Analytics (Students: 29)** UIC
Graduate session (n=25). Overall teaching effectiveness: 4.4; Overall quality of the course: 4.36
Course page

Spring 2021 **CS425: Computer Graphics I (Students: 41)** UIC
Graduate and undergraduate sessions (n=33): Overall teaching effectiveness: 4.03; Overall quality of the course: 4.03
Course page

## Advising & Mentoring (Completed)

Ph.D. theses:

Gustavo Moreira
*Towards an Ecosystem for Visual Analytics Systems Authoring: Insights from the Urban Domain*, 2026
COE Graduate Student Award for Exceptional Research Promise (2026)

Sanjana Srabanti
*Unveiling Inequities: Visualization Frameworks for Multivariate Social Disparities Research*, 2026 (co-advised with G. Elisabeta Marai)

M.Sc. theses:

Vamsi Dath Meka
*Multi-Modal Knowledge Graph Representation of Design Studies for Visual Analytics System Authoring*, 2026

Davide Bartoletti
*Using computer vision for the automatic classification of building facades*, 2023

Venkata Pidaparthi
*Introducing and Enhancing User Provenance in the Curio Framework*, 2025

M.Sc. projects:

Durga Venkata Kashyap Munukutla
*Enhancing Collaborative Urban Analytics in Curio*, 2025

Sajal Chandra
*Data management optimizations for the Urban Toolkit*, 2024

Soham Pradhan
*Transportation Scenario Planning with Graph Neural Networks*, 2022

Others:

B.Sc. projects
Shaun Shannon (2025), Shanmukh Upadhyayula (2025), Ilyan Pang (2025), James Mahia (2025), Priyanjali Rudra (2024), Farah Kamleh (2022), Jayanth Podapati (2022), Daniel Hodczak (2021-2022), Dongjune Park (2026), Andres Quesada (2026), Ho Minh Nhat Nguyen (2026), Dominik Styka (2026)

High-school students
Carissa Chen (2023-2024), Alexandra Orantia (2022-2023)

## Advising & Mentoring (Ongoing)

Ph.D. theses:

Keira Wang (2025-)

Leonardo Ferreira (2023-)
COE Graduate Student Award for Teaching Assistance Excellence (2026)

Kazi Omar (2021-)
*Toward Efficient Computational Frameworks and Modular Urban Visual Analytics Toolkits: Supporting Scenarios and Counterfactuals Reasoning*

Stefan Cobeli (2020-)
COE Graduate Student Award for Teaching Assistance Excellence

## PhD & MSc Committees

**Ph.D. committees**
Nafiul Nipu (UIC, 2026) • Carla Floricel (UIC, 2025) • Mobashir Sadat (UIC, 2024) •Emily Muller (ICL, 2023) • Andrew Wentzel (UIC, 2023) • Carolina Veiga (UFF, 2022) • Shaoyu Chen (NYU, 2022)

**M.Sc. thesis committees**
Raj Paresh Mehta (2025)

**M.Sc. project committees**
Parikshit Solunke (2021) • Pavana Doddi (2021)

**WCP committees**
Stefan Cobeli (2023) • Kazi Omar (2023) • Carla Floricel (2021) • Md Nafiul Alam Nipu (2021) • Andrew Wentzel (2021) • Muhammad Abdul Wahhab (2021)

## Academic Services

**Conference general chair**
• IEEE VIS (2027)

**Program chair**
• SIBGRAPI (2022)

**Conference and workshop organization**
• The 4th Workshop on The Future of Urban Accessibility at ASSETS'24 • 1st Workshop on The Future of Urban Accessibility at ASSETS'22
• VIS 2021, Local co-chair (2021)
• The Future of Global-Scale Spatial Data Collection and Analyses on Urban (in)Accessibility for People with Disabilities Workshop, Co-chair (2021)

**Program committees**

• IEEE VIS Full papers (2022, 2023, 2024, 2026) • IEEE VIS Short papers (2019, 2020, 2021, 2023, 2025)
• EuroVis (2022, 2023, 2024) • Visualization and Data Analysis Conference (2022) • SIBGRAPI (2019, 2020, 2021, 2023)

**Editor**

• Computers & Graphics (SIBGRAPI 2022 Special Issue) • Computers & Graphics (SIBGRAPI 2026 Special Issue)

**Review panels**

• NSF reviewer (2022, 2022, 2024) • Austrian Academy of Sciences (2026)
• Center for Transportation, Equity, Decisions and Dollars reviewer (2022) • Discovery Partners Institute reviewer (2021)

**Journal reviewer**

• IEEE Trans. on Visualization and Computer Graphics (2020, 2021, 2022, 2023, 2024, 2025) • IEEE Trans. on Big Data (2020)
• IEEE Trans. on Intelligent Transportation Systems (2021) • The Visual Computer Journal (2019, 2020, 2021, 2022) • Transportation Research Record Journal (2020) • International Journal of Geo-Information (2021)

**Conference reviewer**

• IEEE VIS (2020, 2021, 2022) • EuroVis (2019, 2020, 2021, 2022) • Sibgrapi (2019, 2020, 2021, 2022)
• VLDB (2021) • WWW (2021) • International Conference on Pattern Recognition (2020, 2021, 2022)

## University Services

**Department committees**

• Graduate Committee (2023-2024, 2024-2025, 2025-2026)
• Faculty Search Committee (2021-2022, 2022-2023)
• Graduate Admission Committee (2020-2021)

**Reviewer**

• Provost's Graduate Research Award reviewer (2020)

## Selected Invited Talks and Presentations

April 2026 — **Towards Extensible and Reproducible Visual Analytics: AI-Guided and Human-Grounded Frameworks for Collaborative System Authoring** — Tampa, FL, USA
Rising Innovators in Scholarly Excellence Series
University of South Florida

October 2025 — **Reinvigorating Environmental Science in Chicago** — Chicago, IL, USA
Beyond100K Summit

March 2025 — **OSCUR: A Platform for Understanding Urban Data - Demonstration and Introduction** — New York City, NY, USA
NYC School of Data 2025

March 2024 — **Interactive Urban Visual Analysis at Scale: From Data to Actionable Insights** — Online
US Census Bureau

March 2023 — **Interactive Urban Visual Analysis at Scale: From Data to Actionable Insights** — Niteroi, RJ, Brazil
Visiting lecturer
Fluminense Federal University

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| December 2022 | **Interactive Visual Analysis of Urban Data: Applications in the Weather and Climate Domains**<br>AGU Fall Meeting 2022<br>American Geophysical Union | Chicago, IL, USA |
| November 2022 | **Mapping Sidewalk Fall Risks Using Big Data and Machine Learning**<br>GSA 2022 Annual Scientific Meeting<br>Gerontological Society of America | Indianapolis, IN, USA |
| October 2022 | **Interactive Visual Analysis at Scale: From Data to Actionable Insights**<br>Keynote speaker<br>SIBGRAPI 2022 | Natal, RN, Brazil |
| July 2022 | **Interactive Visual Analysis of Urban Data: Immersive Analytics Opportunities**<br>Kavli Frontiers of Science Symposium<br>National Academy of Science | Davis, CA, USA |
| April 2022 | **Interactive Visual Analysis of Urban Data: Immersive Analytics Opportunities**<br>The Next Evolution: XR & AEC<br>VRAR CHICAGO | Online |
| January 2022 | **Interactive Visual Analysis of Urban Data: Beyond Flatland**<br>Urban Initiative Program<br>New York University | Online |
| April 2021 | **Interactive Visual Analysis of Urban Data: Beyond Flatland**<br>Department of Energy Computer Graphics Forum 2021<br>Department of Energy | Online |
| April 2021 | **Interactive Visual Analysis of Urban Data: A Computational Perspective on Cities**<br>Federal University of Rio Grande do Sul | Porto Alegre, RS, Brazil |
| March 2021 | **Interactive Visual Analysis of Urban Data: A Computational Perspective on Cities**<br>Fluminense Federal University | Niteroi, RJ, Brazil |
| December 2018 | **Exploration of Street-Level Images at Scale**<br>Pedestrian Movement Technology Showcase at Metro North | New York City, NY, USA |
| November 2018 | **Shadow Accrual Maps: Efficient Accumulation of City-Scale Shadows over Time**<br>IEEE Visualization Conference (VIS) | Berlin, Germany |
| June 2018 | **Time Lattice: A Data Structure for the Interactive Visual Analysis of Large Time Series**<br>EG/VGTC Conference on Visualization (EuroVis) | Brno, Czech Republic |
| October 2017 | **TopKube: A Rank-Aware Data Cube for Real-Time Exploration of Spatiotemporal Datasets**<br>IEEE Visualization Conference (VIS) | Phoenix, AZ, USA |
| September 2016 | **Visualizing and Exploring Urban Data**<br>Data Visualization Summit | Boston, MA, USA |
| October 2016 | **TopKube: A Rank-Aware Data Cube for Real-Time Exploration of Spatiotemporal Datasets**<br>Data Systems for Interactive Analysis Workshop (DSIA) | Chicago, IL, USA |

## Professional Memberships

Association for Computing Machinery (ACM)